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1  # Note 16: RandomizedSearchCV & GridSearchCV
2  # =====
3
4  # Import Libraries
5  from sklearn.model_selection import GridSearchCV
6  from sklearn.ensemble import RandomForestClassifier
7  from sklearn.model_selection import RandomizedSearchCV
8  from sklearn.ensemble import RandomForestClassifier
9
10
11  # 01. Grid Serach CV with Random Forest
12  # =====
13
14  # Create a Model
15  model = RandomForestClassifier(random_state=42)
16
17  # Set Values
18  param_grid = {
19      'n_estimators': [50, 100, 200],
20      'max_depth': [3, 5, 10]
21  }
22
23  # Run Grid Search CV
24  grid = GridSearchCV(model, param_grid, cv=5, scoring='accuracy')
25
26  # Fit Model
27  grid.fit(X_train, y_train)
28
29  # Print Data
30  print(grid.best_params_)
31  print(grid.best_score_)
32
33  # 02. Grid Serach CV with SVM
34  # =====
35
36  parameters = {
37      'C': [1, 10, 100],
38      'kernel': ['linear', 'poly', 'rbf']
39  }
40
41  gsv = GridSearchCV(svm, parameters)
42  gsv.fit(X_train, y_train)
43
44  print("Best C Value For GSV : ", gsv.best_score_)
45  print("Best Parameters      : ", gsv.best_params_)
46
47  # 03. Randomized Serach CV with SVM
48  # =====
49
50  parameters = {
51      'C': [1, 10, 100],
52      'kernel': ['linear', 'poly', 'rbf']
53  }
54
55  rsv = RandomizedSearchCV(svm, parameters, n_iter=5)
56  rsv.fit(X_train, y_train)
57  print("Best C Value For RSV : ", rsv.best_score_)
58  print("Best Parameters      : ", rsv.best_params_)
59
60  # 04. Randomized Serach CV with Random Forest
61  # =====
62
63  model = RandomForestClassifier(random_state=42)
64
65  param_grid = {
66      'n_estimators': [50, 100, 200, 300],
67      'max_depth': [3, 5, 10, 15],
68      'min_samples_split': [2, 5, 10]
69  }
70
71  random_search = RandomizedSearchCV(
72      model,

```

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73     param_grid,  
74     n_iter=10,  
75     cv=5,  
76     scoring='accuracy',  
77     random_state=42  
78 )  
79  
80 random_search.fit(X_train, y_train)  
81  
82 print(random_search.best_params_)  
83 print(random_search.best_score_)
```